

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.kernel_ridge import KernelRidge
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

auto-mpg.csv

auto-mpg.csv(text/csv) - 18131 bytes, last modified: 8/17/2026 - 100% done
Saving auto-mpg.csv to auto-mpg.csv

```
df = pd.read_csv("auto-mpg.csv")

df["horsepower"] = pd.to_numeric(
    df["horsepower"], errors="coerce"
)

df = df.dropna(
    subset=["horsepower", "mpg"]
)

print("Dataset Shape:", df.shape)
display(df.head())
```

Dataset Shape: (392, 9)

	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	origin	car name
0	18.0	8	307.0	130.0	3504	12.0	70	1	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693	11.5	70	1	buick skylark 320
2	18.0	8	318.0	150.0	3436	11.0	70	1	plymouth satellite
3	16.0	8	304.0	150.0	3433	12.0	70	1	amc rebel sst
4	17.0	8	302.0	140.0	3449	10.5	70	1	ford torino

```
X = df["horsepower"].values.astype(float)
y = df["mpg"].values.astype(float)

print("Number of records:", len(X))
print("Feature: Horsepower")
print("Target: MPG")
```

Number of records: 392
Feature: Horsepower
Target: MPG

```
def locally_weighted_regression(x_query, X, y, tau=25):

    weights = np.exp(
```

```

        -((X - x_query) ** 2) / (2 * tau ** 2)
    )

    A = np.column_stack(
        (np.ones(len(X)), X)
    )

    W = np.diag(weights)

    theta = np.linalg.pinv(
        A.T @ W @ A
    ) @ A.T @ W @ y

    return theta[0] + theta[1] * x_query

X_test = np.linspace(
    X.min(),
    X.max(),
    200
)

lwr_predictions = np.array([
    locally_weighted_regression(
        x, X, y
    )
    for x in X_test
])

print("LWR prediction completed.")

```

LWR prediction completed.

```

rbf_model = KernelRidge(
    alpha=0.5,
    kernel="rbf",
    gamma=0.002
)

rbf_model.fit(
    X.reshape(-1, 1),
    y
)

rbf_predictions = rbf_model.predict(
    X_test.reshape(-1, 1)
)

print("RBF prediction completed.")

```

RBF prediction completed.

```

plt.figure(figsize=(10, 6))

plt.scatter(

```

```
X,  
y,  
s=15,  
label="Real Auto MPG Data"  
)  
  
plt.plot(  
    X_test,  
    lwr_predictions,  
    linewidth=2,  
    label="Locally Weighted Regression"  
)  
  
plt.plot(  
    X_test,  
    rbf_predictions,  
    linewidth=2,  
    label="RBF Regression"  
)  
  
plt.xlabel("Horsepower")  
plt.ylabel("Miles Per Gallon (MPG)")  
plt.title(  
    "Locally Weighted Regression vs RBF"  
)  
  
plt.legend()  
plt.grid()  
plt.show()
```

Locally Weighted Regression vs RBF

```
print("Comparison")
print("-----")
print("LWR : Local, query-dependent model")
print("RBF : Smooth radial-basis model")
print("Dataset : Real Auto MPG Kaggle Dataset")
```

